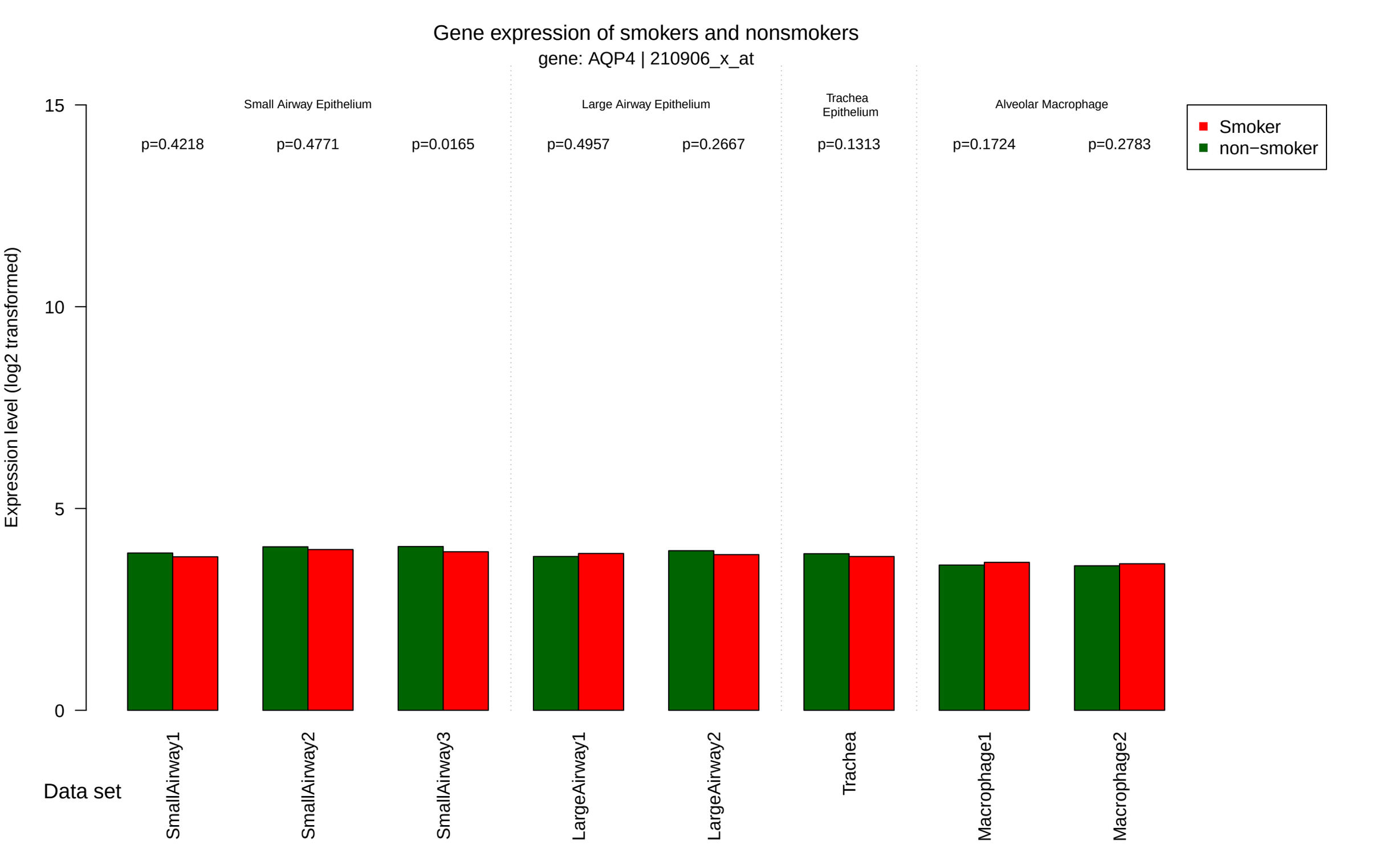
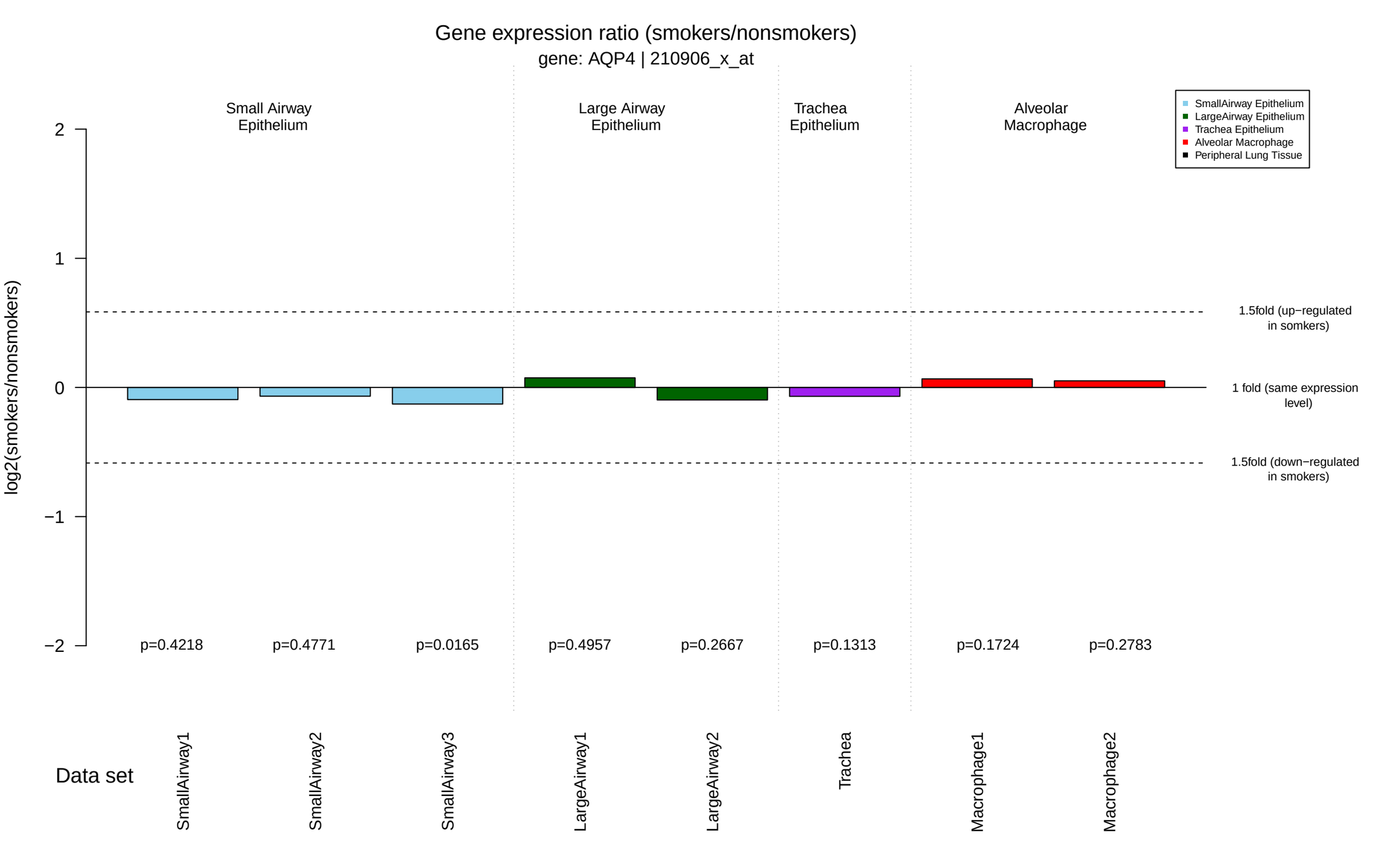
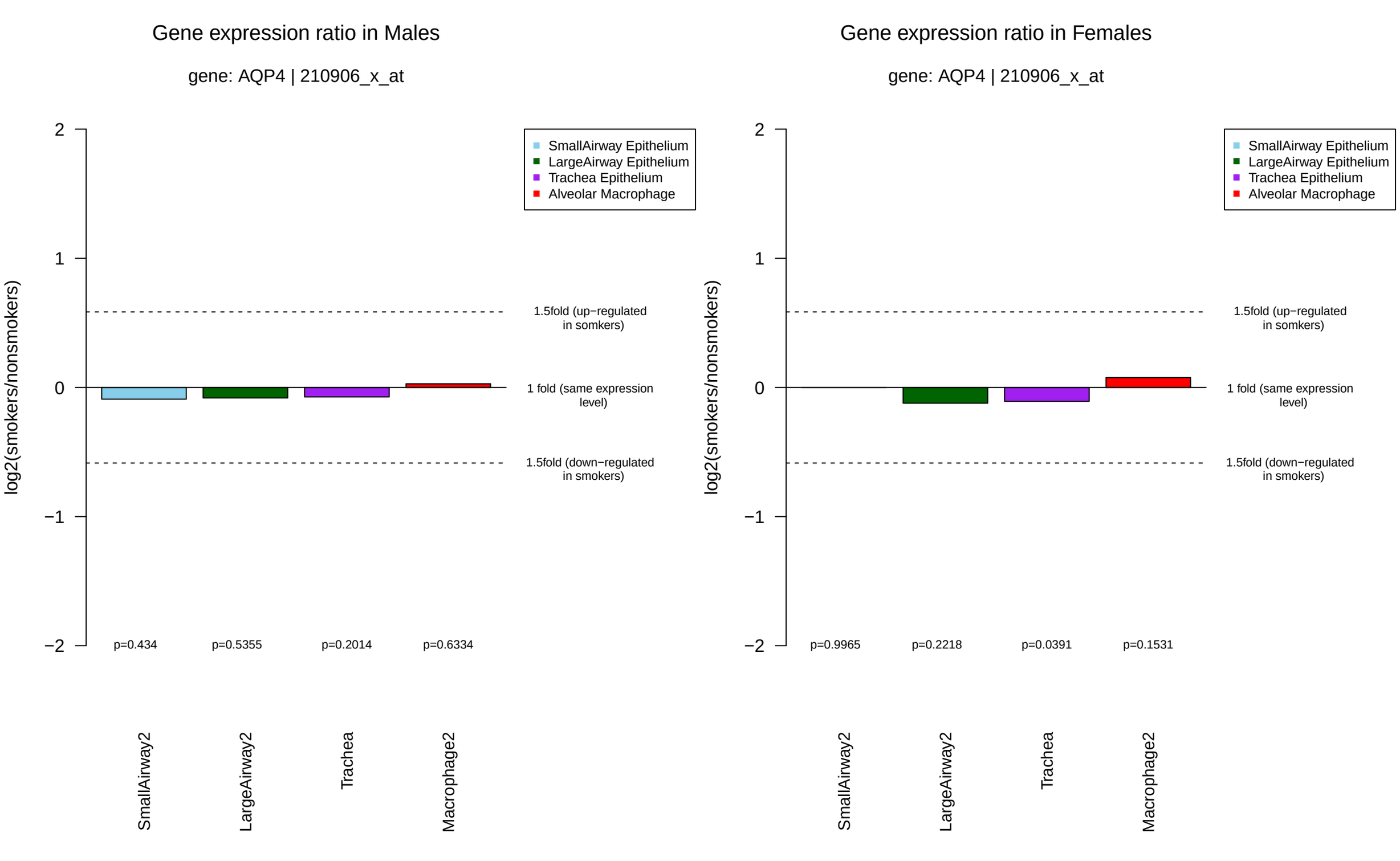
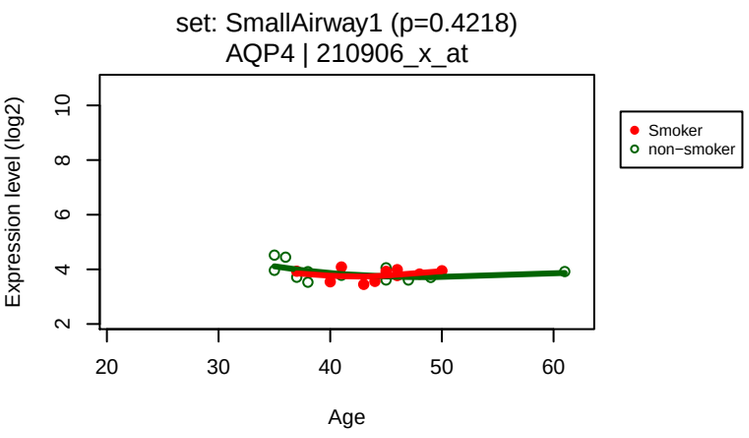
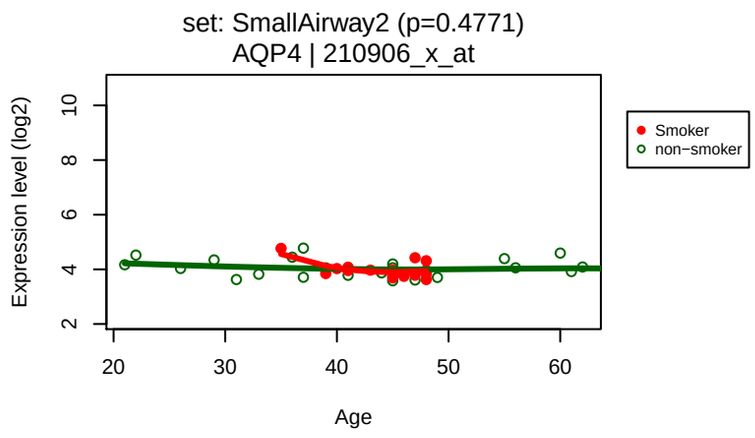
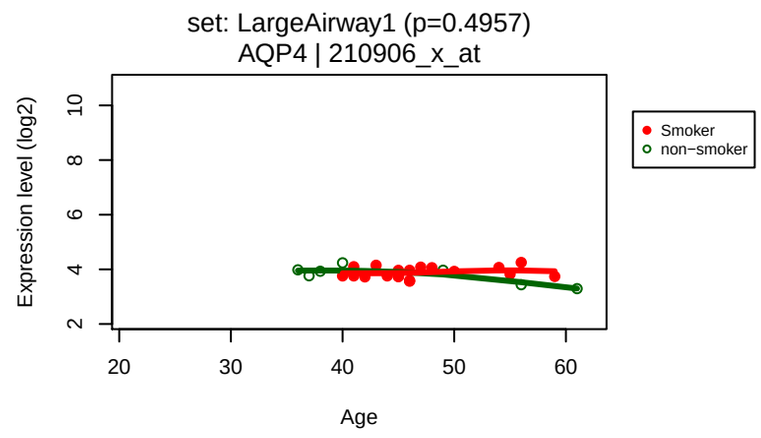

Gene expression of smokers and nonsmokers
gene: AQP4 | 210906_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4218
p=0.4771
p=0.0165
p=0.4957
p=0.2667
p=0.1313
p=0.1724
p=0.2783
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

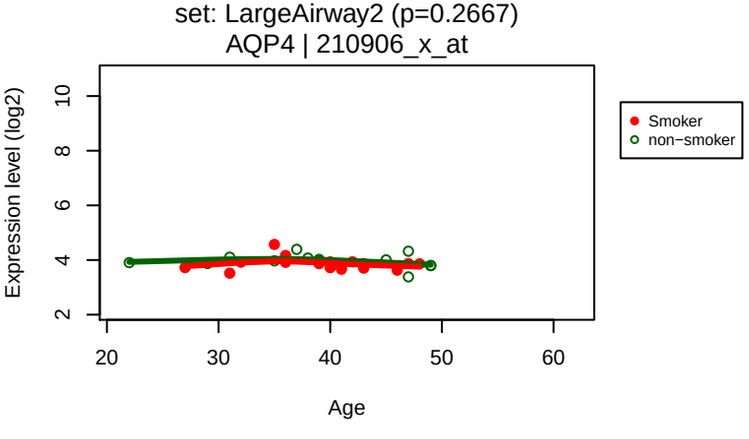

Gene expression ratio (smokers/nonsmokers)
gene: AQP4 | 210906_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4218
p=0.4771
p=0.0165
p=0.4957
p=0.2667
p=0.1313
p=0.1724
p=0.2783
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

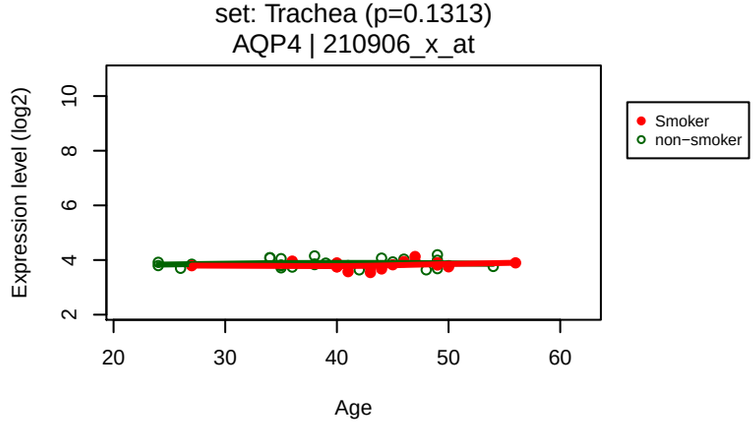

Gene expression ratio in Males
gene: AQP4 | 210906_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.434
p=0.5355
p=0.2014
p=0.6334
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AQP4 | 210906_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9965
p=0.2218
p=0.0391
p=0.1531
SmallAirway2
LargeAirway2
Trachea
Macrophage2

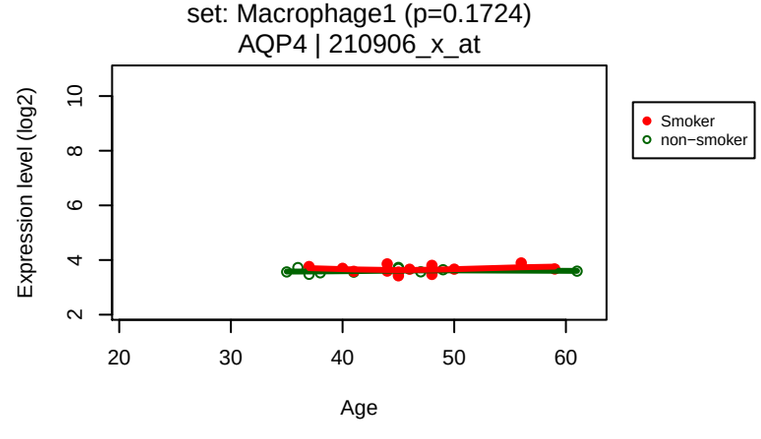

set: SmallAirway1 (p=0.4218)
AQP4 | 210906_x_at
Expression level (log2)
Age
Smoker
non−smoker

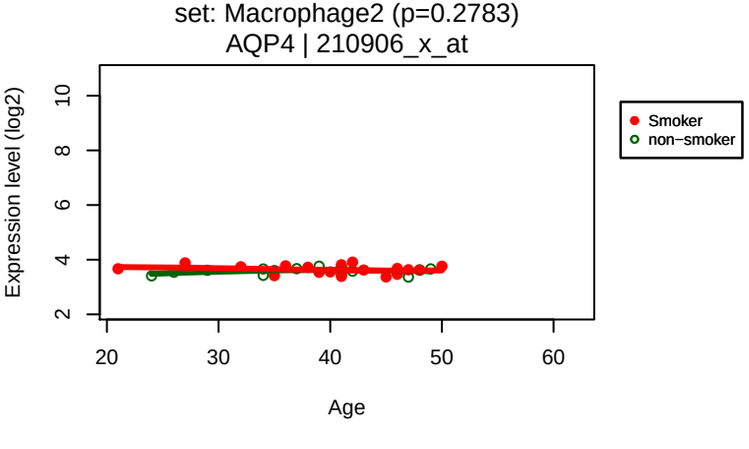

set: SmallAirway2 (p=0.4771)
AQP4 | 210906_x_at
Expression level (log2)
Age
Smoker
non−smoker

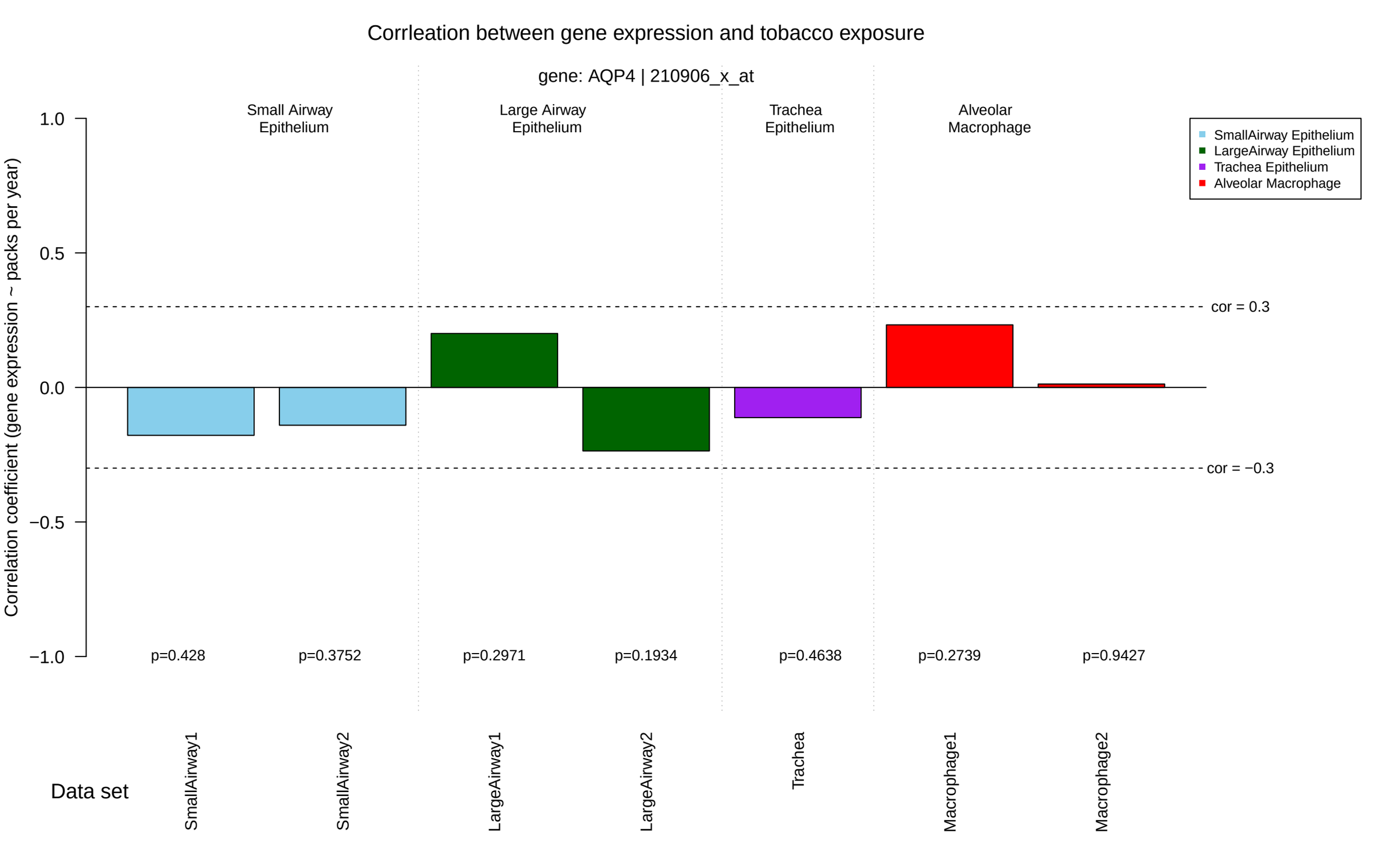

set: LargeAirway1 (p=0.4957)
AQP4 | 210906_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2667)
AQP4 | 210906_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1313)
AQP4 | 210906_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1724)
AQP4 | 210906_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2783)
AQP4 | 210906_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AQP4 | 210906_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.428
p=0.3752
p=0.2971
p=0.1934
p=0.4638
p=0.2739
p=0.9427
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2